

Roll No. 2582300041

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ZE-195

**B.Tech. (New Scheme) (B.E.):
(Main/ATKT) Examination, 2026**

(Fourth Semester)

SIGNAL AND SYSTEMS

EL-404

Time : 3 Hours]

[Maximum Marks : 60

Note : Attempt all questions. All questions carry equal marks.

1. (a) Explain the different types of systems.
- (b) Explain convolution integral and convolution sum.

Or

- (a) Explain continuous time signal and Discrete time signal.
- (b) Explain causal and non-causal system.

2. (a) Explain continuous time Fourier Transform.
(b) State and prove sampling theorem.

Or

- (a) State the properties of Fourier series.
(b) State and explain Parseval's theorem.
3. (a) Define Laplace Transform.
(b) Explain state space model.

Or

- (a) Explain the concept of ROC.
(b) Explain concept of controllability and observability.
4. (a) What do you mean by Z-Transform ?
(b) Explain phase delay.

Or

- (a) Explain the properties of Z-Transform.
(b) Explain group delay.

5. (a) Explain Random process in detail.
(b) Explain power spectral density.

Or

- (a) Explain continuous and Discrete Random variables.
(b) Explain the Wiener-Khinchin-Einstein theorem.

